

AC Line Conversion to DC

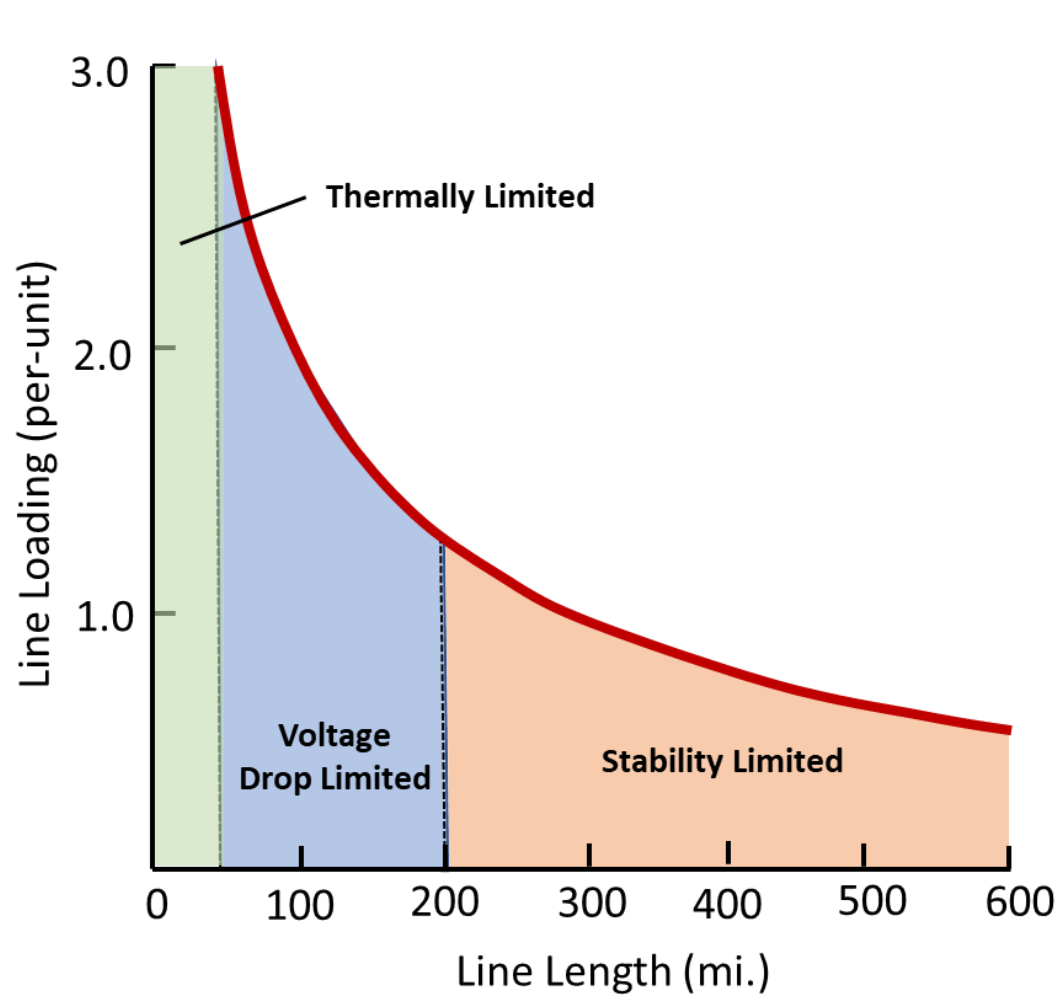


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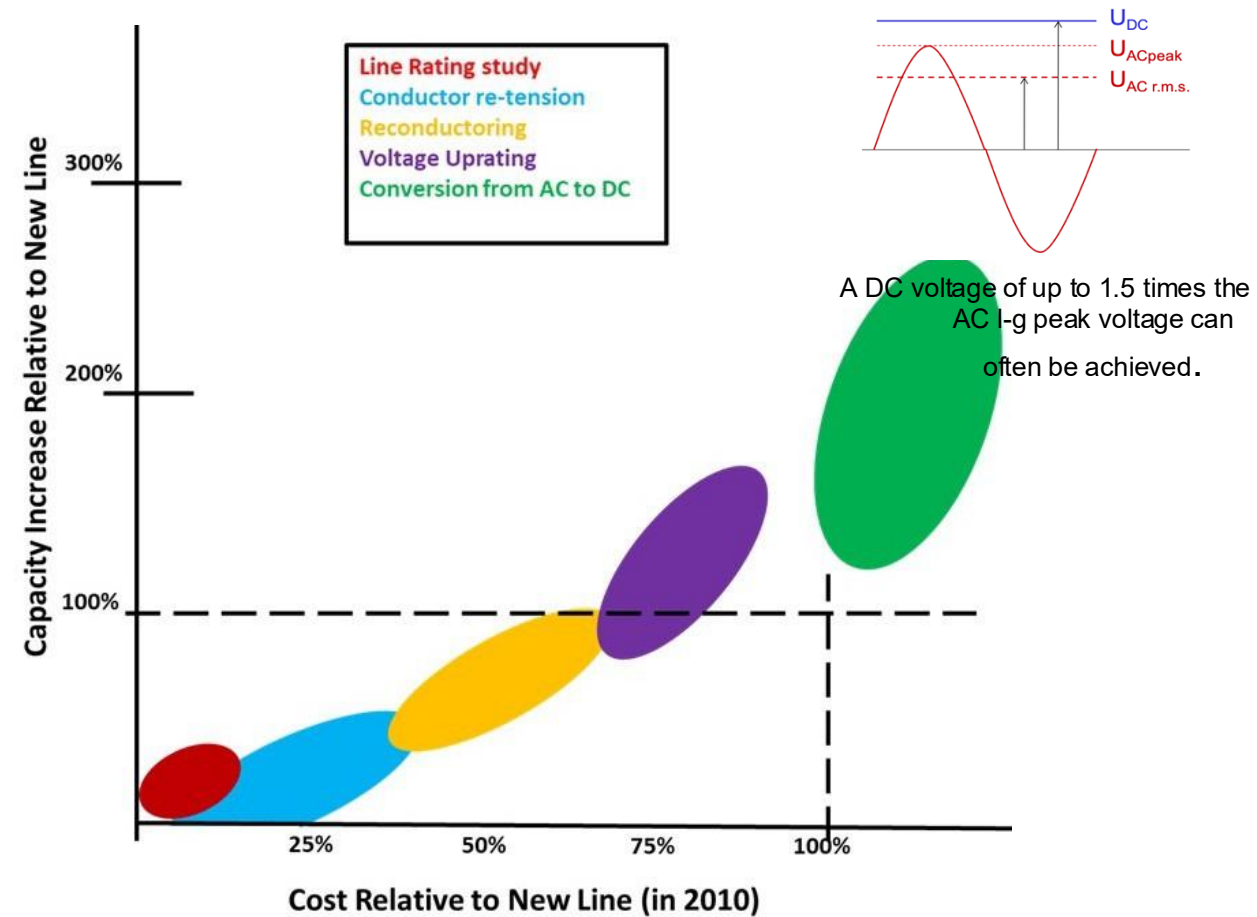
HVDC Tech Transfer for PG&E

April 30, 2026

Increasing Capacity of Existing OH Transmission Lines



Universal Loadability Curve of ac Lines



Potential capacity increase vs. relative cost

What does AC to DC Conversion Mean / Implications

Structures – No change

Conductors – No change

Insulators – Re-insulation may be required

Evaluation of Electric Field Effects

Terminal substations – DC converter stations must be built

Re-permitting for DC



Line Configuration Options

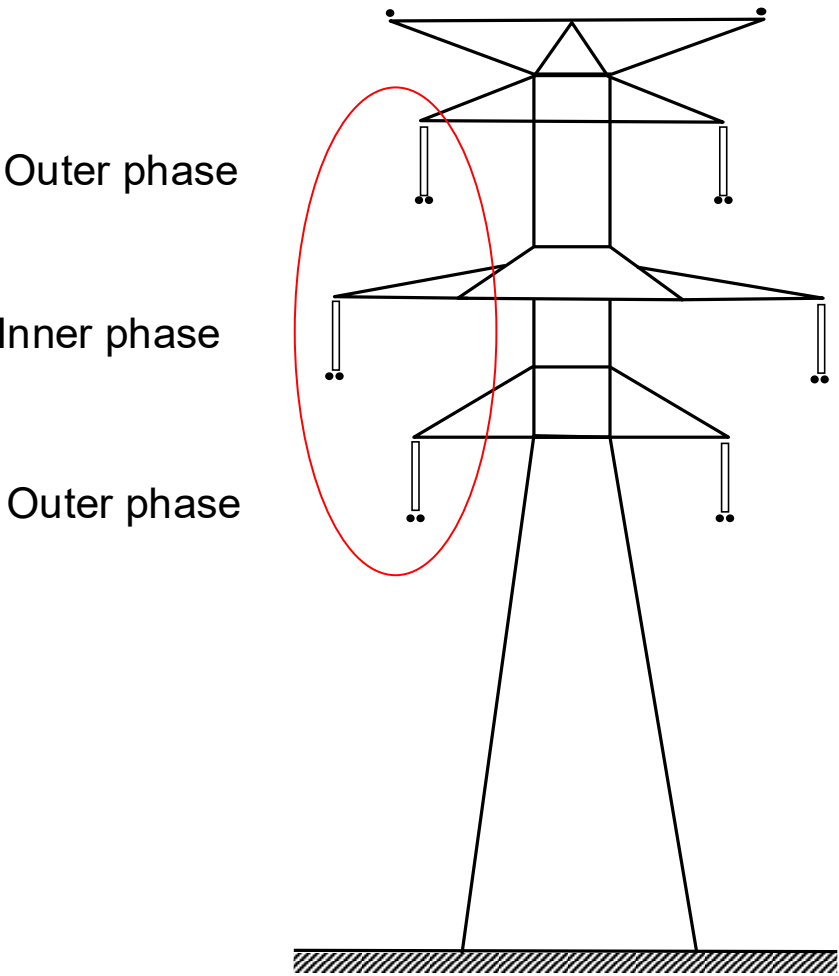
Monopole • + • + • +	Bipole • + • 0 • -
Tripole • + • +/- • -	Hybrid ~ • • + ~ • • 0 ~ • • -

Single Circuit

- Outer phases used as a traditional Bipole – centre phase used as emergency metallic ground return
- Tripole configuration

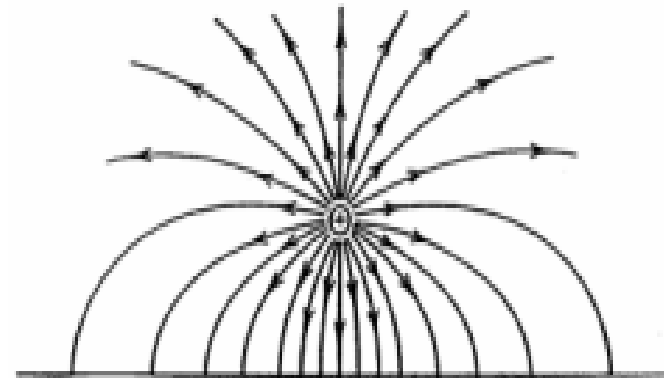
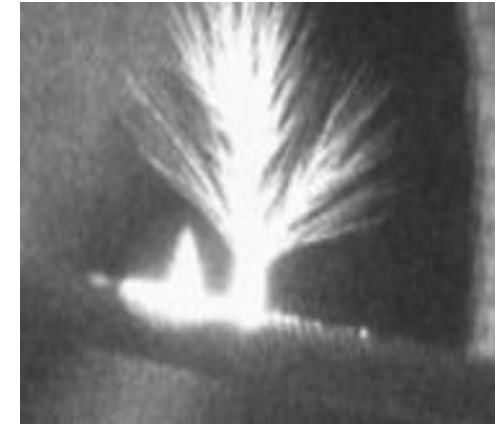
Double Circuit

- One circuit converted – as above
- Three conventional Bipoles
- A single Bipole (3 bundles making up a single pole)
- Two Tripoles



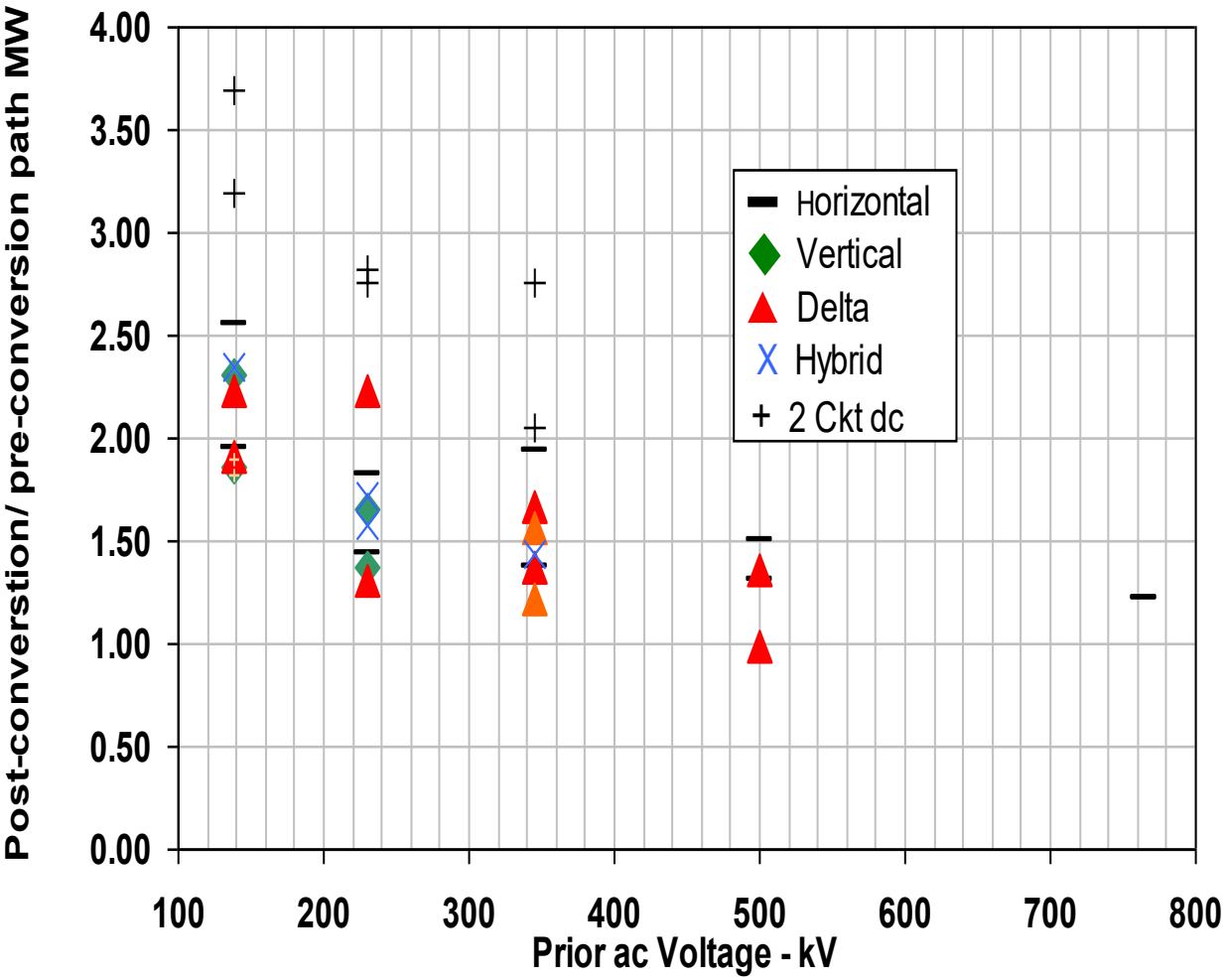
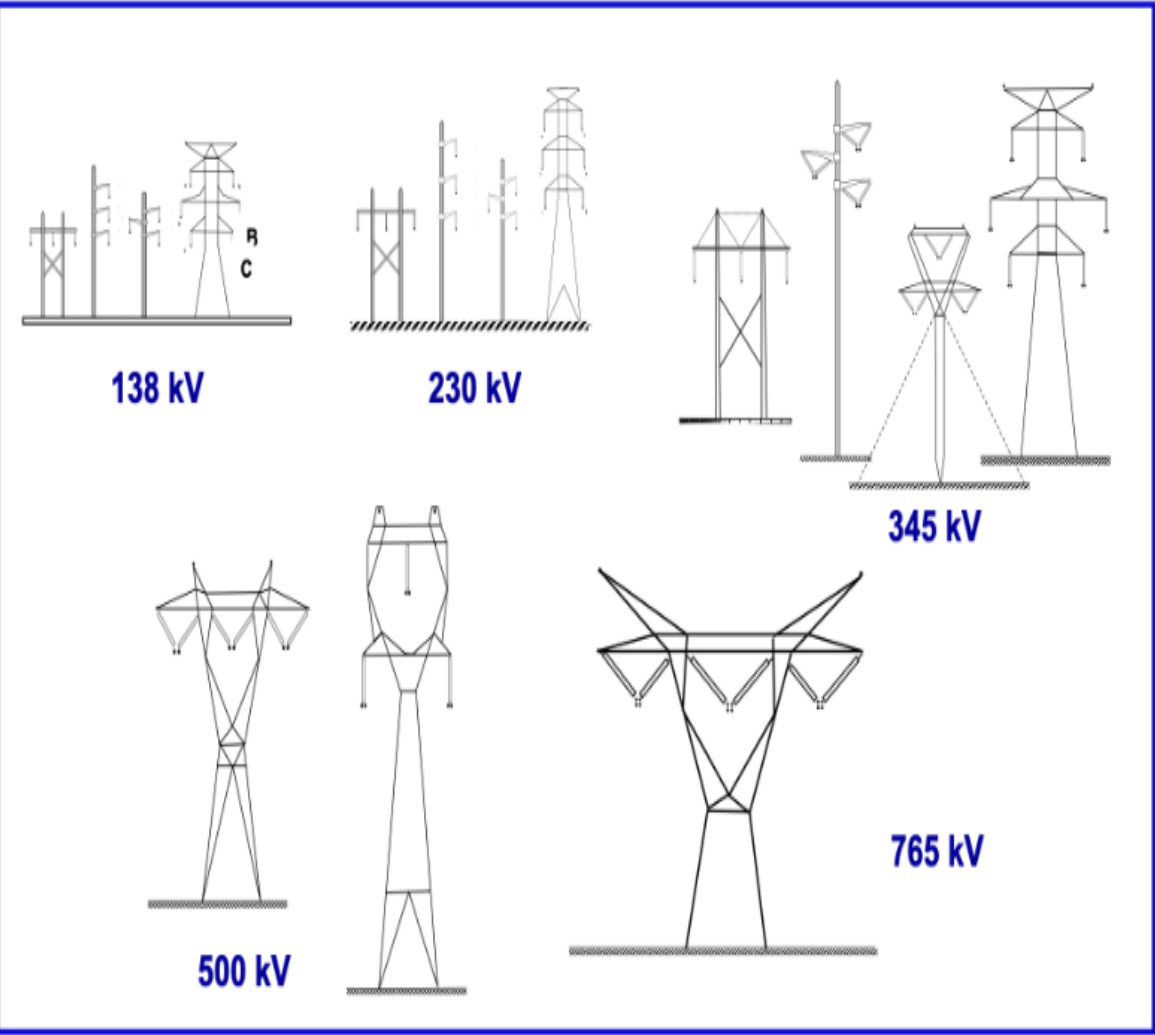
AC to DC Conversion: Key Considerations

- Conductor gradient
 - Conductor & Hardware Corona
 - Audible Noise
- Ground-level electric field
- Insulator Contamination
- Clearance for insulators at the structure
- NESC clearance to ground
- Live Working Clearances



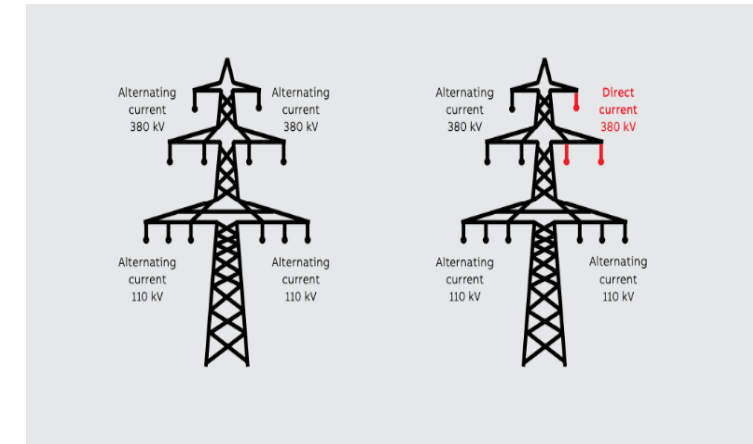
Requires Studies & Testing to Confirm Performance

North American Transmission Structures & Possible Capacity Gains



Recent AC to DC Conversion -One circuit of a Double Circuit 380 kV AC line in Germany being converted to DC – Hybrid AC / DC on Same Tower

- UltraNet project in Germany
- Line runs between North Rhine Westphalia and Baden-Württemberg
- Line Length 340 km
- Two 380 kV & Two 110 kV AC circuits on the same tower
- One 380 kV AC circuit was converted to ± 380 kV DC
- Facilitate transmission of about 2,000 MW (**about 40% increase in capacity**) from wind farms in the North Sea to the industrial towns in the south of Germany.
- Other AC lines have also been studied with a view to conversion.





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